

Conjugated Systems & Aromaticity

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1 Aromaticity & Aromatic Substitution

1.1 Aromatic Compounds

Core Concept 1.1: Aromatic Compounds

The name "aromatic" originates from these compounds' characteristic scent; however, the term now denotes a strictly structural definition, as covered in Core Concept 1.5.

While aromaticity is not limited to the base benzene structure, these notes will primarily refer to benzene-derived compounds such as the following.

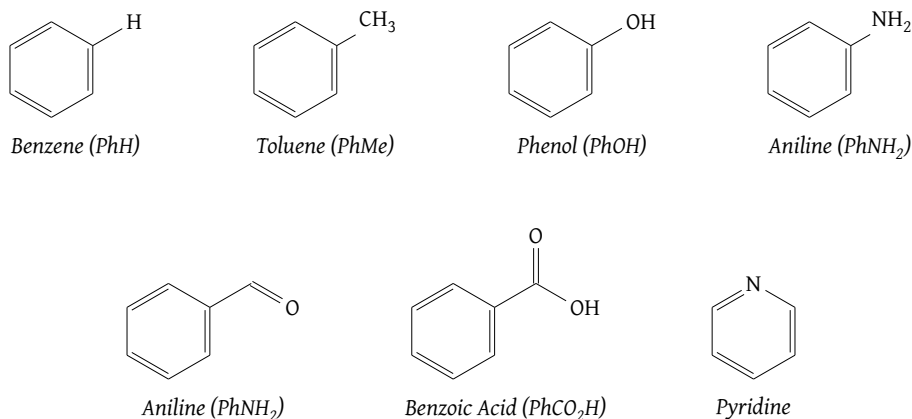


Figure 1: Phenyl ring-based aromatic compound examples; pyridine represents an aromatic compound with a non-CH atom as part of its ring

Core Concept 1.2: Aromatic Compound Stereochemistry

Additional substituent positions on the root aromatic compounds shown above can be specified using the ortho, meta, and para keywords.

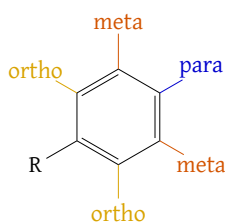


Figure 2: Phenyl ring substituent-positioning keywords

The compound naming scheme follows the structure *relation-substituent-root*. Consider the following examples.

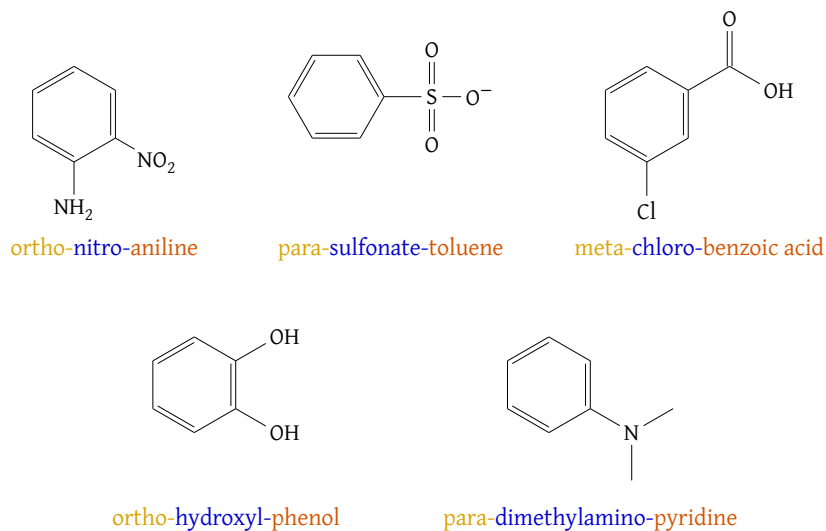


Figure 3: Phenyl ring-based aromatic compound nomenclature

Core Concept 1.3: Reactivity of Aromatic Compounds

Aromatic compounds are considerably less reactive than would otherwise be predicted. For instance, while a simple cyclohexene reacts readily with mCPBA to form an epoxide, no reaction occurs when benzene is mixed with three equivalents of mCPBA.

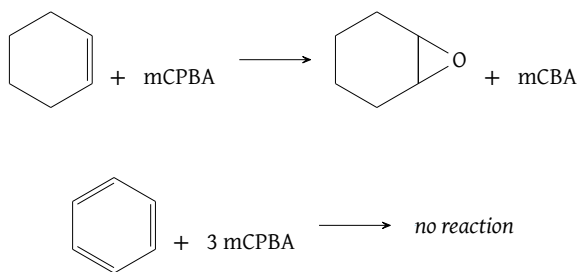


Figure 4: Reactions of cyclohexene and benzene with mCPBA

Similarly, a substantial portion of the predicted energy release during the hydrogenation of benzene is offset by aromatic stabilization.

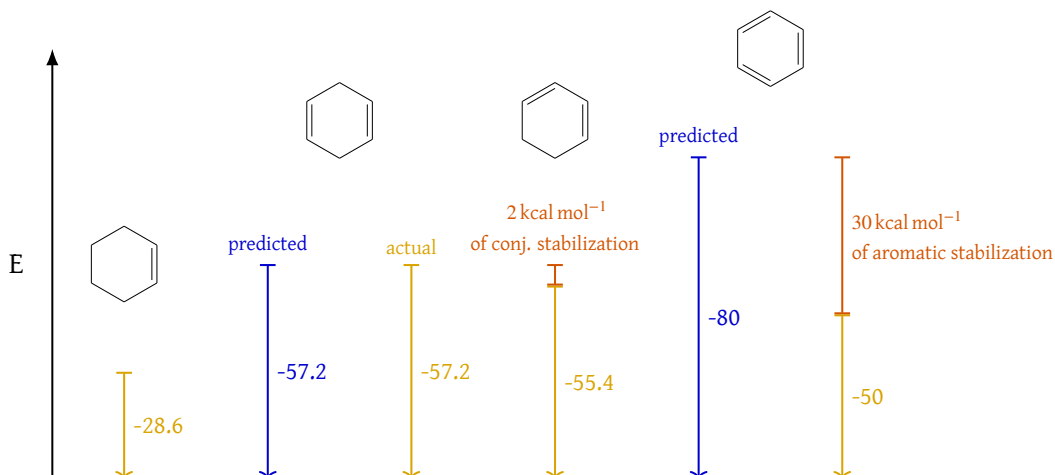


Figure 5: Energy diagram of predicted versus actual heats of hydrogenation for cyclohexene derivatives; aromaticity confers a substantial increase in stability relative to the summed values of the corresponding isolated double-bond hydrogenations, exceeding even the stabilization observed in nonaromatic but conjugated cyclohexene systems

To explain this phenomenon, consider a top-down view of the benzene π molecular orbital diagram.

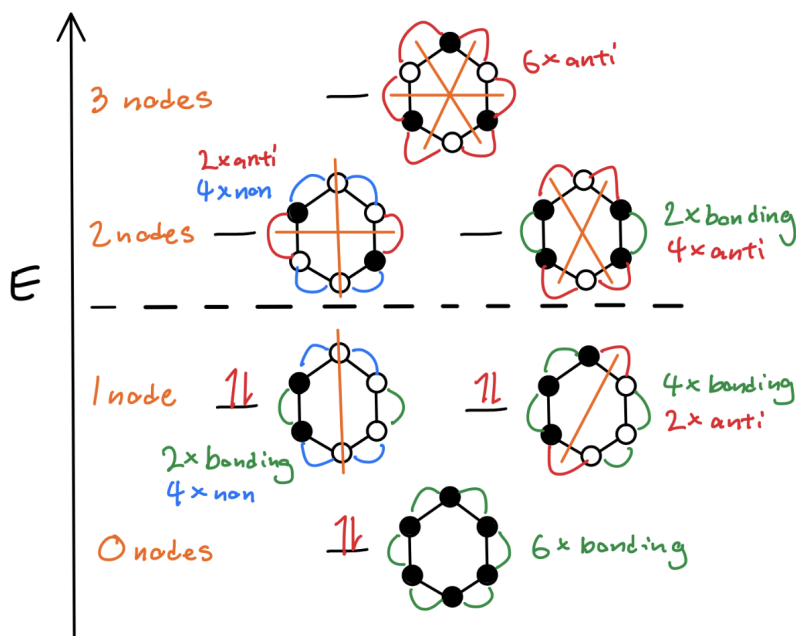


Figure 6: Energy diagram of the benzene π molecular orbitals (top-down view)

As shown in Figure 6, the π electrons occupy exclusively bonding molecular orbitals, rendering the system exceptionally stable.

Core Concept 1.4: Aromatic Compound Molecular Diagrams

The **Frost Diagram** is a method for rapidly predicting π molecular orbital energy levels of a cyclic conjugated system, following the steps below.

- (1) The compound must possess a cyclic array of p orbitals
- (2) Draw the ring structure with one vertex pointed downwards
- (3) Draw a circle around the compound
- (4) Intersections represent energy levels
- (5) The middle of the circle represents the nonbonding energy level
- (6) Add e^- s

Below are the corresponding orbital diagrams for a cyclopentenyl anion and a cycloheptenyl cation; the benzene diagram has already been shown above.

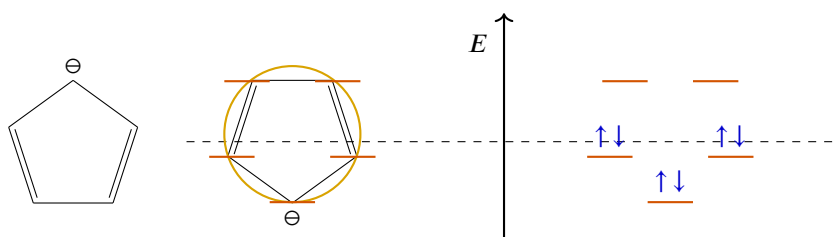
(a) Cyclopentene Anion

Figure 7: Frost diagram for the cyclopentenyl anion, showing three filled bonding/nonbonding orbitals and two unoccupied, degenerate antibonding orbitals

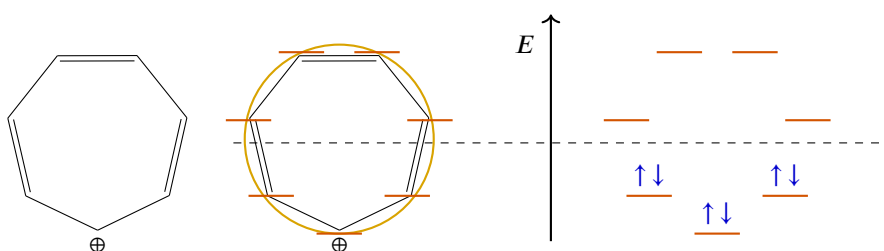
(b) Cycloheptene Cation

Figure 8: Frost diagram for the cycloheptenyl cation, showing three filled bonding/nonbonding orbitals and four unoccupied, degenerate antibonding orbitals

Core Concept 1.5: Aromaticity Rules

In order for a compound to be considered aromatic, it must satisfy a strict set of rules.

- (1) The compound must be a cyclic array of p orbitals
- (2) The e^- count of the cyclic array must fulfill the equation of $4n + 2$, where n is an integer (0, 1, 2, 3, 4, ...).

*An e^- count that instead fulfills the equation $4n$, where n is an integer, results in a compound that is far more reactive than predicted and is termed **anti-aromatic**.*

If the above criteria are not met, the compound is considered **nonaromatic**, as in the cyclohexene shown below.

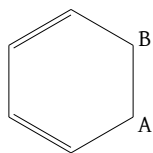


Figure 9: Nonaromatic cyclohexene; carbons A and B are sp^3 , which prevents an uninterrupted cyclic array of p orbitals

As practice, consider the following two examples.

(a) **Pentene Anion**

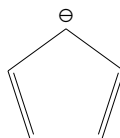


Figure 10: Aromatic pentene anion

This structure exhibits a cyclic array of p orbitals with an electron count of 6, satisfying the $4n + 2$ expression. This compound is therefore **considered aromatic**.

(b) **Pentene Cation**

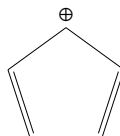


Figure 11: Anti-aromatic pentene cation

This compound likewise exhibits a cyclic array of p orbitals, but its electron count of 4 satisfies $4n$ rather than $4n + 2$. This compound is therefore considered **anti-aromatic**.

Notice that the corresponding molecular orbital diagram results in two unpaired electrons, accounting for its heightened reactivity.

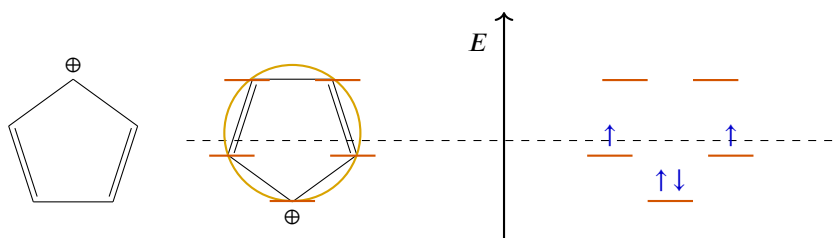


Figure 12: Frost diagram for the cyclopentenyl cation

Core Concept 1.6: Aromatic Contributors

An **aromatic contributor** refers to a resonance structure of a molecule that possesses a fully conjugated, cyclic, planar π system satisfying the $4n + 2$ π -electron count, even when the dominant or lowest-energy Lewis

structure does not immediately appear aromatic. Consider the following molecule as an example.

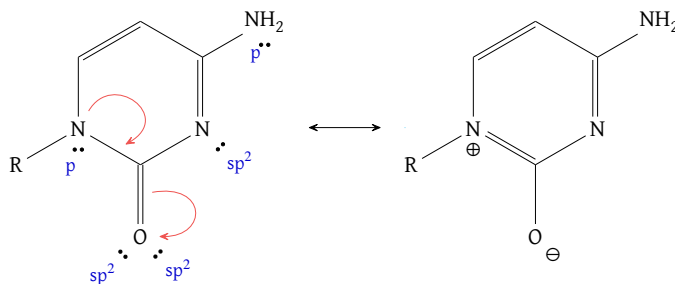


Figure 13: Aromatic contributor; a simple rule of thumb for identifying a resonance contributor is to attempt to achieve alternating π bonds while ensuring that a circle drawn around the ring never crosses a π bond

Notice that although the base resonance structure is nonaromatic by definition, since its conjugated π orbital system is not cyclic, an alternative structure accessible through resonance is aromatic, thereby qualifying the molecule as possessing an aromatic contributor.

1.2 Electrophilic Aromatic Substitution

Core Concept 1.7: Electrophilic Aromatic Substitution

Electrophilic aromatic substitution (EAS) describes a class of reactions in which an aromatic ring functions as the nucleophilic species. The general mechanism underlying this reaction type is outlined below.

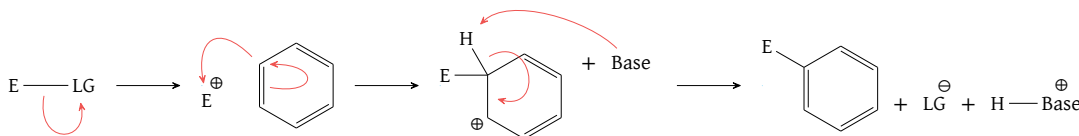


Figure 14: General mechanistic outline of electrophilic aromatic substitution reactions

1.2.1 Reaction Types

Core Concept 1.8: Halogenation

EAS halogenation results in the attachment of a single halogen atom to an aromatic ring, with the corresponding hydrohalic acid generated as a byproduct. This mechanism proceeds via activation by a **Lewis acid** (defined as any chemical species capable of accepting an electron pair), as illustrated below.

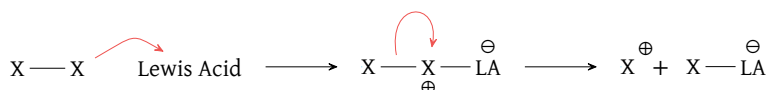


Figure 15: Generation of a halogen cation electrophile via Lewis acid activation; Lewis acids function as electron-pair acceptors and include species such as FeX_3 , BX_3 , and AlX_3

The chlorination and bromination of benzene, presented below, illustrate this general mechanism.

(a) Chlorination of Benzene

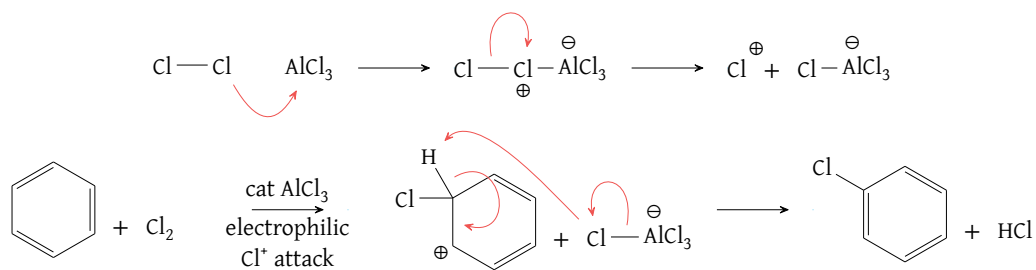


Figure 16: Chlorination of benzene

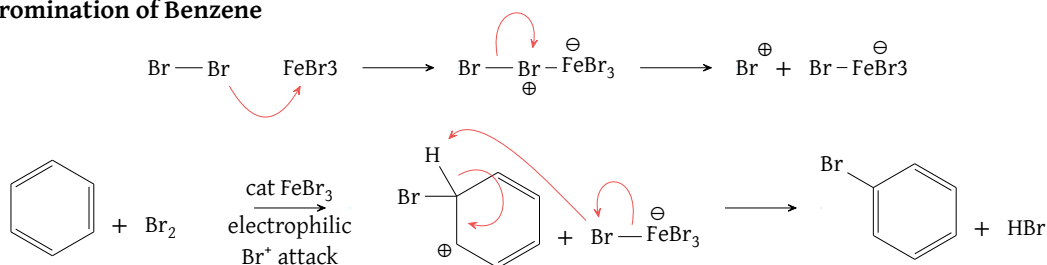
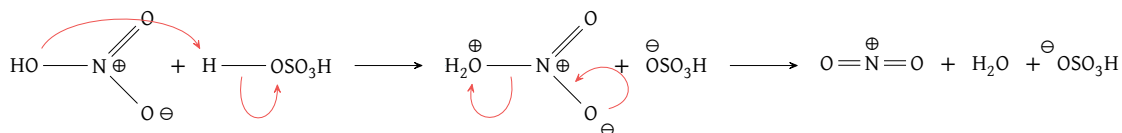
(b) Bromination of Benzene

Figure 17: Bromination of benzene

Core Concept 1.9: Nitration

Nitration involves the attachment of a nitro group (NO_2) to an aromatic ring and serves primarily as a synthetic intermediate step in practice. Consider first the generation of the NO_2^+ electrophile, formed upon exposure of nitric acid to sulfuric acid.

Figure 18: Generation of the NO_2^+ electrophile

With this electrophile established, the nitration of benzene proceeds according to the general mechanism outlined in Core Concept 1.7.

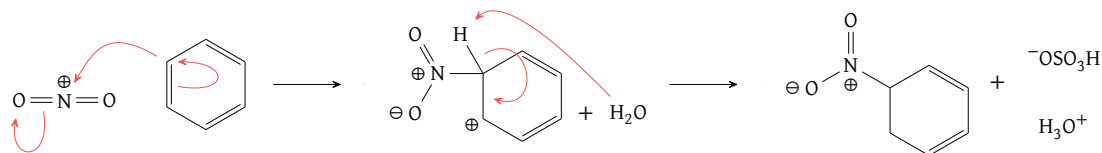
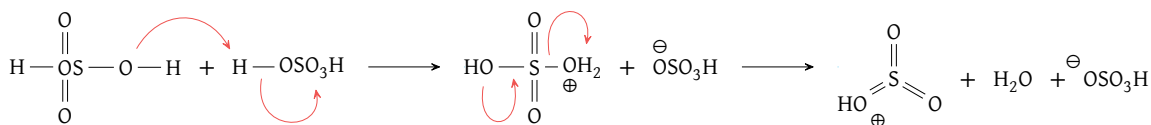


Figure 19: Nitration of benzene

Core Concept 1.10: Sulfonation

Sulfonation entails the attachment of a sulfonic acid group (SO_3H) to an aromatic ring. The initial step involves a reaction between two molecules of H_2SO_4 , arising from the use of **fuming H_2SO_4** as one of the reagents.

Figure 20: Formation of fuming H_2SO_4

The reaction then proceeds according to the same general EAS mechanism described in Core Concept 1.7.

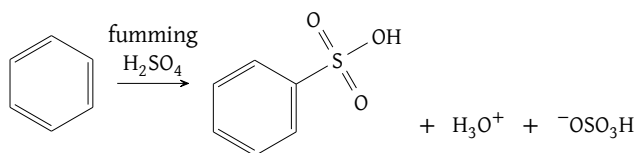
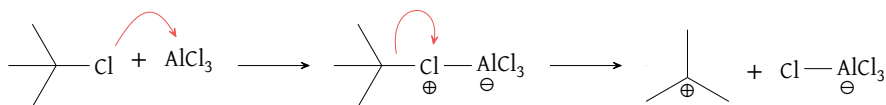


Figure 21: Sulfonation of benzene

Core Concept 1.11: Alkylation

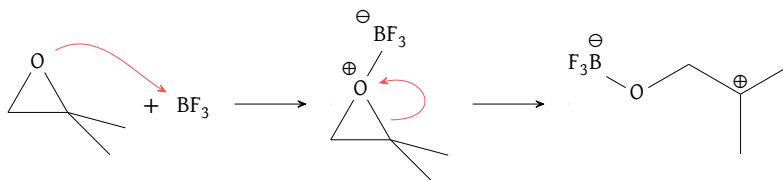
Alkylation refers to the addition of an alkyl group to an aromatic structure. Before an alkane can act as an electrophile, it must first undergo conversion to a carbocation intermediate, as shown in the reactions below.

(a) Generation of an Electrophile via a Lewis Acid

Figure 22: Mechanism for the formation of a tert-butyl carbocation from tert-butyl chloride using aluminum chloride (AlCl_3) as a Lewis acid catalyst

This transformation parallels the initial step of the chlorination pathway presented in Core Concept 1.8.

(b) Lewis Acid-Catalyzed Epoxide Ring Opening

Figure 23: Boron trifluoride (BF_3) coordination to an epoxide, promoting regioselective ring cleavage to generate the more thermodynamically stable tertiary carbocation

(c) Acid-Catalyzed Carbocation Formation from an Alcohol

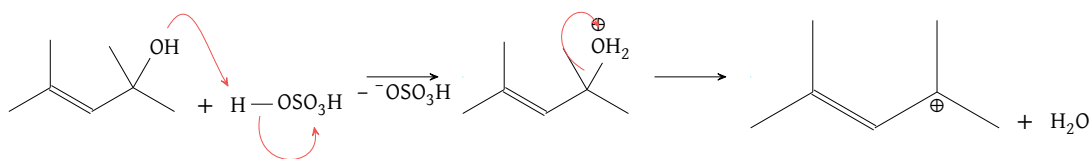
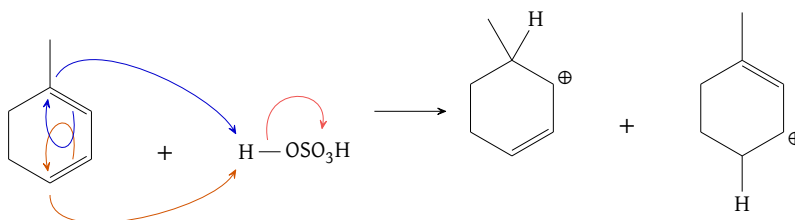


Figure 24: Protonation of a tertiary alcohol by sulfuric acid, followed by the expulsion of water to form a tertiary carbocation intermediate

(d) Protonation of an Aromatic Ring

Figure 25: Electrophilic attack on toluene by H^+ ; the first hydrogenation step follows either the blue or orange e^- movement

In certain cases, carbocation rearrangement occurs when the initially formed carbocation is unstable.

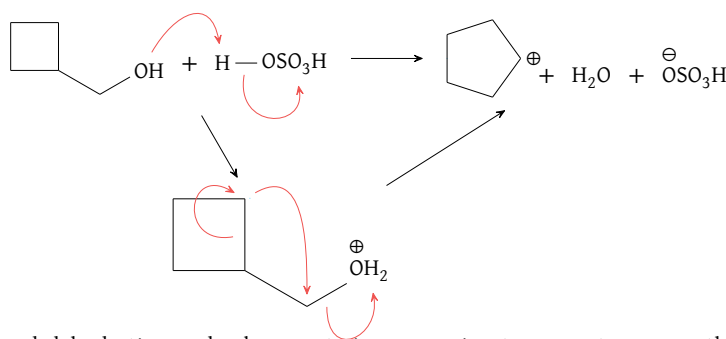


Figure 26: Acid-catalyzed dehydration and subsequent ring expansion to generate a more thermodynamically stable cyclopentyl carbocation

These rearranged carbocations subsequently undergo EAS with aromatic structures.

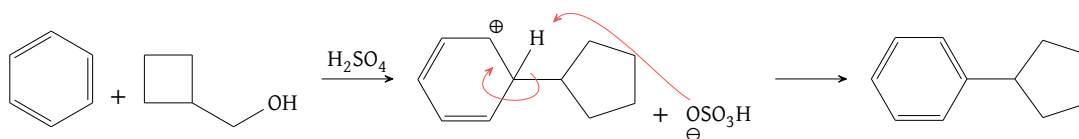


Figure 27: Electrophilic aromatic substitution of benzene by a rearranged cyclopentyl carbocation

In the case above, illustrating rearrangement from a cyclobutyl to a cyclopentyl carbocation, such rearrangement can be avoided through a stepwise pathway involving **acylation** (Core Concept 1.12) followed by reduction with Zn(Hg) and HCl .

(a) Acylation

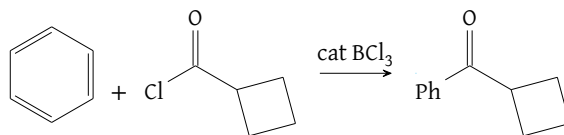


Figure 28: Acylation

(b) Reduction

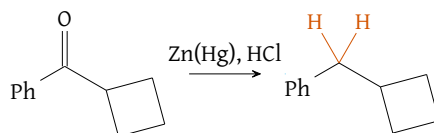


Figure 29: Reduction of ketone with Zn(Hg) and HCl

Core Concept 1.12: Acylation

Acylation refers to the addition of an acyl halide to an aromatic structure. The acyl halide must first be rendered electrophilic through activation by a Lewis acid, as shown in the general reaction outline below.

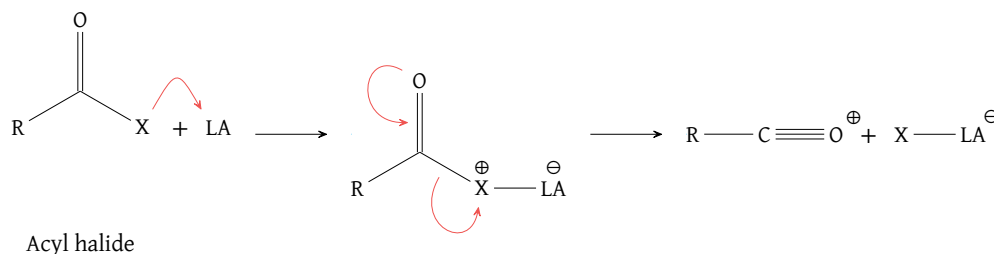


Figure 30: Conversion of an acyl halide to an electrophile

The resulting RCO^+ species can then undergo EAS with an aromatic ring, as demonstrated below using representative compounds.

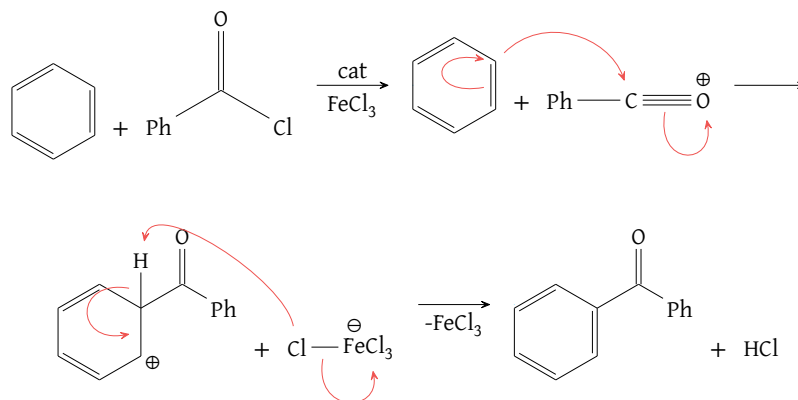


Figure 31: Acylation of benzene

1.2.2 Directing Groups**Core Concept 1.13: Directing Groups**

So far, we have covered the EAS reaction types in Subsubsection 1.2.1 using benzene as our sole relevant aromatic structure. However, in scenarios where the aromatic ring is asymmetrical, chemists require a method for determining which carbon the EAS reaction will occur on. We now introduce **directing groups**, functional groups already attached to a benzene ring that dictate where (*ortho*, *meta*, *para*) an incoming electrophile will land.

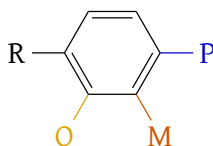


Figure 32: Directing groups ortho (yellow), meta (orange), and para (blue)

The table below summarizes directing group behavior; consult each Core Concept entry for a more detailed treatment of each type.

Directing Group Type	Examples	Description
(1) Inductive O/P Director	$R = \text{alkyl [Me, Et, } i\text{Pr, ...]}$	Alkyl groups donate electron density inductively, directing substitution to the ortho and para positions.
(2) Inductive M Director	$R = \text{carbon w/ EN atoms [CCl}_3, \text{CF}_3, \text{...}]}$	Carbons bearing electronegative atoms withdraw electron density inductively, directing substitution to the meta position.
(3) Resonance O/P Director	i) R has lone pair adjacent to ring: [OMe, SR, NMe ₂] ii) $R = \text{carbon w/ } \pi \text{ system: [alkene, alkyne, benzene]}$	Substituents that donate electron density via resonance into the ring direct substitution to the ortho and para positions.
(4) Resonance M Director	[C=O, C $\equiv$ N, NO ₂]	Substituents that withdraw electron density via resonance from the ring direct substitution to the meta position.

Table 1: Classification of electrophilic aromatic substitution directing groups

Core Concept 1.14: Inductive O/P Directors

Inductive O/P directors are substituents that direct any EAS to the ortho and para positions within an aromatic ring.

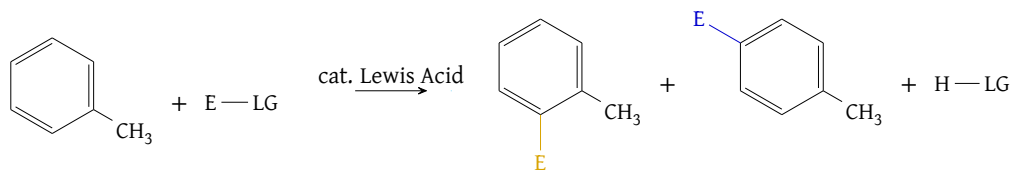
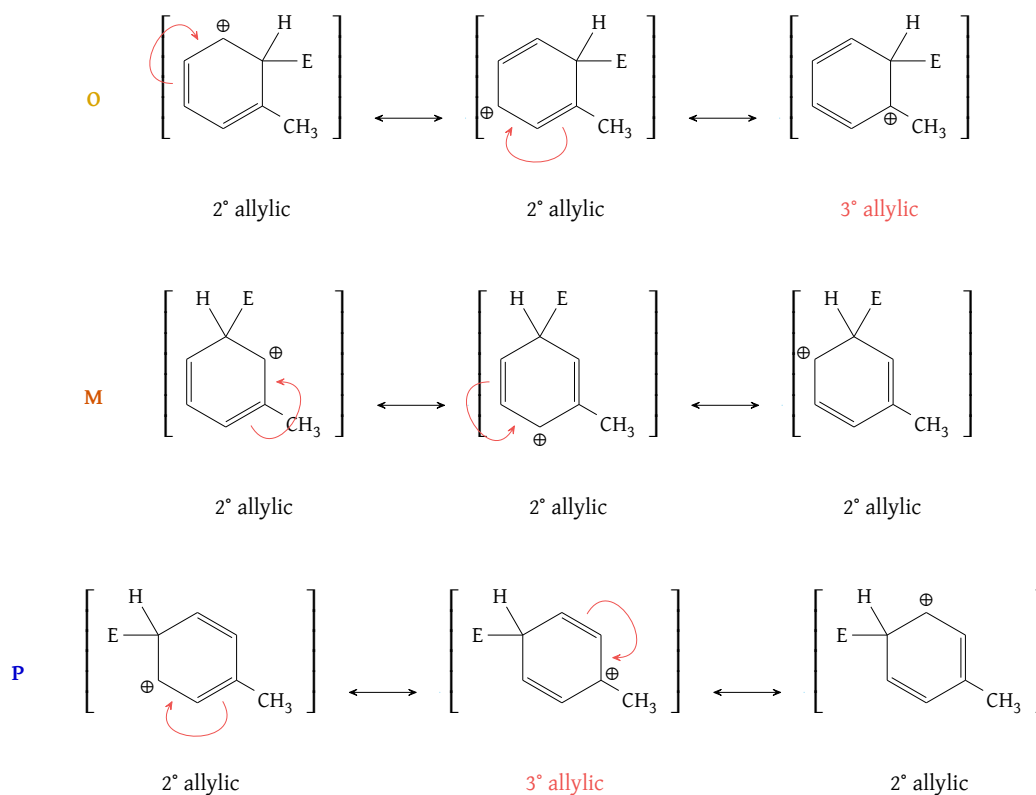
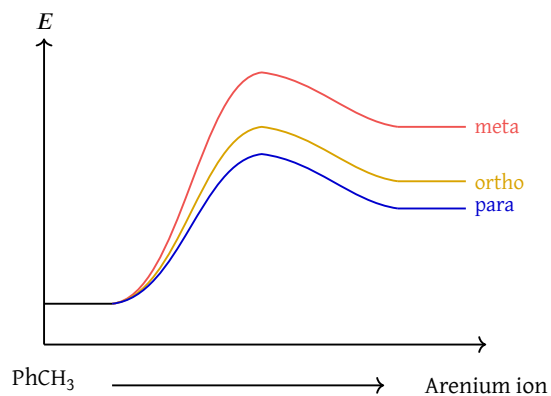


Figure 33: Inductive O/P directors

These substituents are typically alkyl groups; their preference for O/P over M positioning arises from the relative stability of the resonance structures generated in each case.

Figure 34: Ortho, meta, para positioning resonance structures of H_3CPh

As shown above in Figure 34, although most resonance structures place the transient carbocation charge at a secondary allylic position, O/P substitution permits a tertiary allylic carbocation in one of three resonance structures, whereas M substitution does not. Consequently, the relative energies of the resulting carbocation intermediates follow the ordering shown in the energy diagram below.

Figure 35: Energy diagram comparing the relative energies of the meta, ortho, and para transition states leading from the reaction $\text{H}_3\text{CPh} + \text{E}^+ \longrightarrow \text{H}_3\text{CPhE}^+$

Core Concept 1.15: Inductive M Directors

Inductive M directors are substituents that direct EAS to the meta position within an aromatic ring. These substituents consist of carbons bonded to highly electronegative atoms, rendering the carbon strongly partially positive.

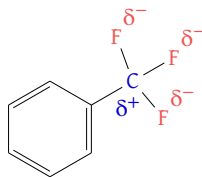


Figure 36: Inductive M directors

The M position is favored because the O/P 3° allylic carbocation resonance structures experience destabilizing repulsion between the carbocation and the partially positive substituent carbon. The M position, by contrast, permits a resonance structure in which the positive substituent carbon and the carbocation charge reside at maximal distance on opposite sides of the aromatic ring.

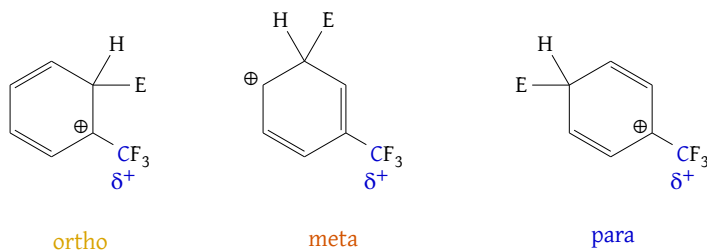


Figure 37: Best resonance contributors of each position in F_3CPh ; the O/P 3° allylic contributors experience additional Coulombic **repulsion** due to the partially positive adjacent carbon of the CF_3 substituent

Consequently, the resulting energy levels place O/P positioning above that of M positioning.

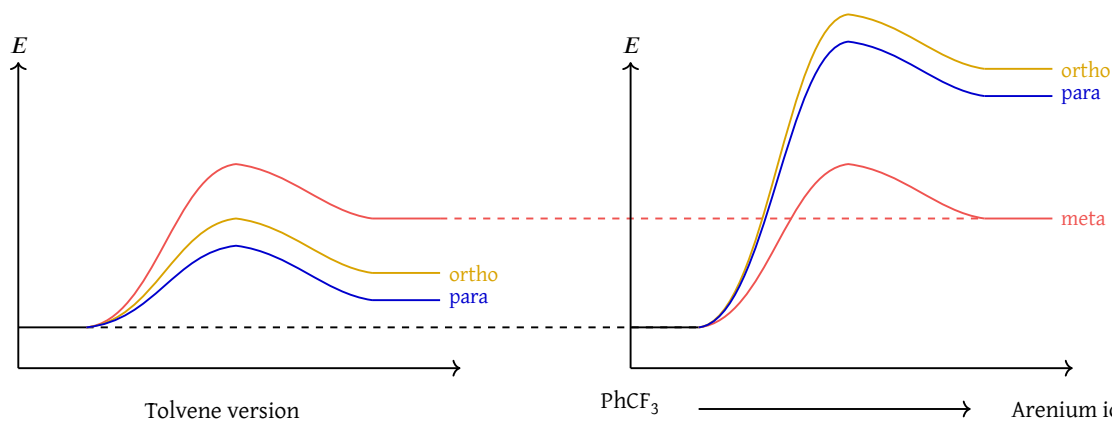


Figure 38: Energy diagram comparing the transition-state energies for meta, ortho, and para substitution in the reactions $H_3CPh + E^+ \longrightarrow H_3CPhE^+$ and $F_3CPh + E^+ \longrightarrow F_3CPhE^+$; inductive directors raise the O/P energies comparatively more, making meta the relatively favored position

Core Concept 1.16: Resonance O/P Directors

Resonance O/P directors are substituents that direct EAS to the O and P positions within an aromatic ring by donating e^- density. Phenol, for example, possesses resonance structures that localize the resulting carbanion charge exclusively at the O and P positions.

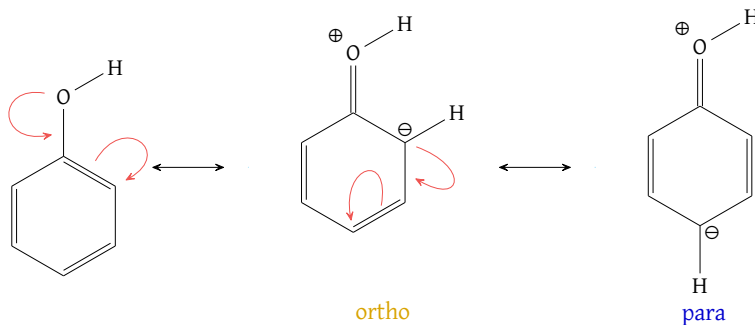


Figure 39: Resonance O/P director; this mechanism requires a substituent capable of donating an electron pair

These substituents must specifically possess a lone pair adjacent to the aromatic ring. Another substituent type capable of e^- donation is a carbon bearing a conjugated π system, as illustrated by the following alkene.

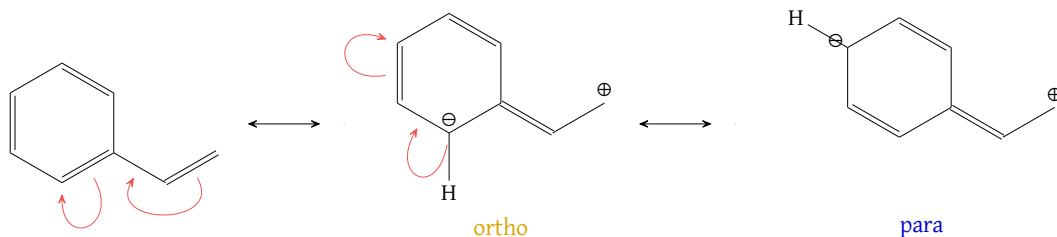


Figure 40: Conjugated π system resonance O/P directing

Note that formation of the primary carbocation is favorable here because it is allylic.

As a side note on how inductive and resonance directors differ in the mechanism by which they induce positional favorability: resonance directors alter the relative nucleophilicity of the ring positions, whereas inductive directors influence favorability through transition-state stability.

Core Concept 1.17: Resonance M Directors

Resonance M directors are substituents that direct EAS to the M position within an aromatic ring by withdrawing e^- density. Carbonyl substituents, for example, withdraw e^- density such that the resulting resonance structures place a positive charge at the O/P positions, rendering these positions less nucleophilic and the M position relatively more nucleophilic.

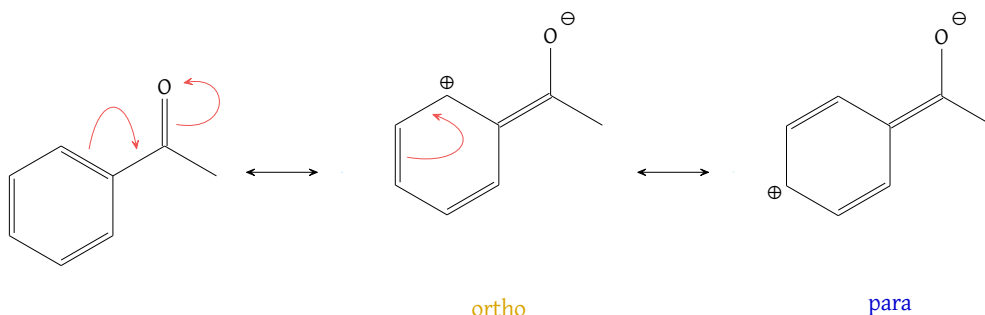


Figure 41: Resonance M director

As a result, when an electrophile approaches the transient aromatic ring, nucleophilic attack occurs most favorably from the M position rather than the O/P positions.

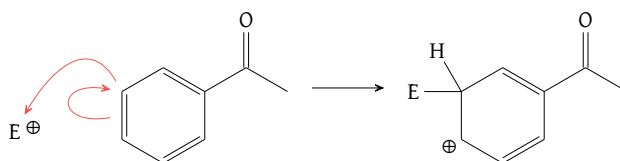


Figure 42: Electrophilic aromatic substitution in the presence of resonance M directors

Core Concept 1.18: Halogens as Directing Groups

Halogens can also function as directing groups; empirically, they act as O/P directors. To understand the basis for this behavior, consider monohalobenzene in the context of both the inductive effect and resonance.

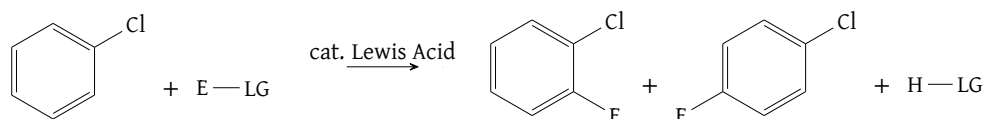


Figure 43: Halogen directing groups

(a) Inductive Effect

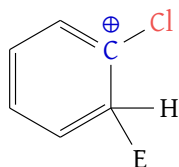


Figure 44: Inductive effect

Under an O/P directing EAS interaction, inductive analysis would actually disfavor O/P substitution, since the resulting carbocation transition state places positive charge adjacent to the electronegative halogen, further depleting an already electron-poor center. The resulting dipole destabilizes the structure.

(b) Resonance

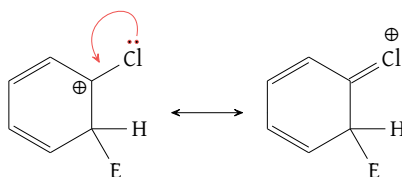


Figure 45: Resonance capability of halogen directing groups

Resonance, by contrast, allows the halogen to donate e^- density to satisfy the octet requirement throughout the molecule, favoring the O/P directing interaction.

Although these two effects act in opposition, experimental data confirm that resonance dominates over the inductive effect, resulting in O/P substitution being favored in practice.

1.2.3 Aromatic Substituent Manipulations

Core Concept 1.19: Reduction of Acyl Ketone

The reduction of an acyl ketone, previously employed in Core Concept 1.11, replaces a carbonyl oxygen with two hydrogen substituents.

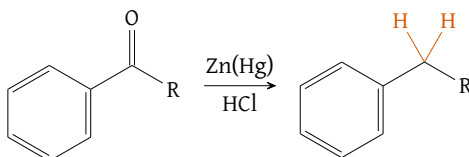


Figure 46: Reduction of an acyl ketone with Zn(Hg) and HCl

No mechanism is provided for this reaction.

Core Concept 1.20: Nitrobenzene Reduction to Aniline

The reduction of nitrobenzene proceeds analogously to the reduction of an acyl ketone (Core Concept 1.19), in which oxygens are replaced by hydrogens, and employs the same reactants to accomplish this transformation.

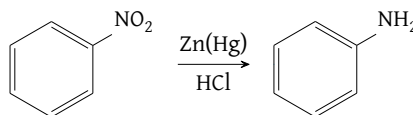


Figure 47: Nitrobenzene reduction to aniline with Zn(Hg) and HCl

No mechanism is provided for this reaction.

Core Concept 1.21: Diazotization of Aniline

The diazotization of aniline is a natural subsequent step following nitrobenzene reduction (Core Concept 1.20), as the product of the latter serves as the reactant for the former.

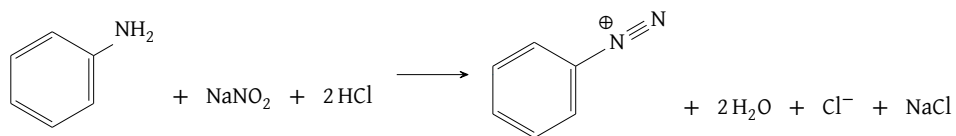
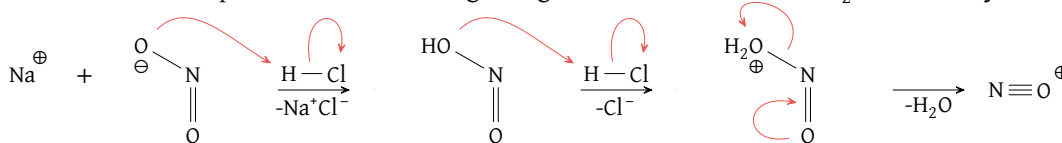


Figure 48: Diazotization of aniline

The reaction mechanism proceeds as follows, beginning with the conversion of NO_2^- to a nitrosyl cation.

Figure 49: Conversion of NO_2^- to a nitrosyl cation

The highly reactive nitrosyl cation then interacts with aniline as shown below to produce **aryl diazonium**.

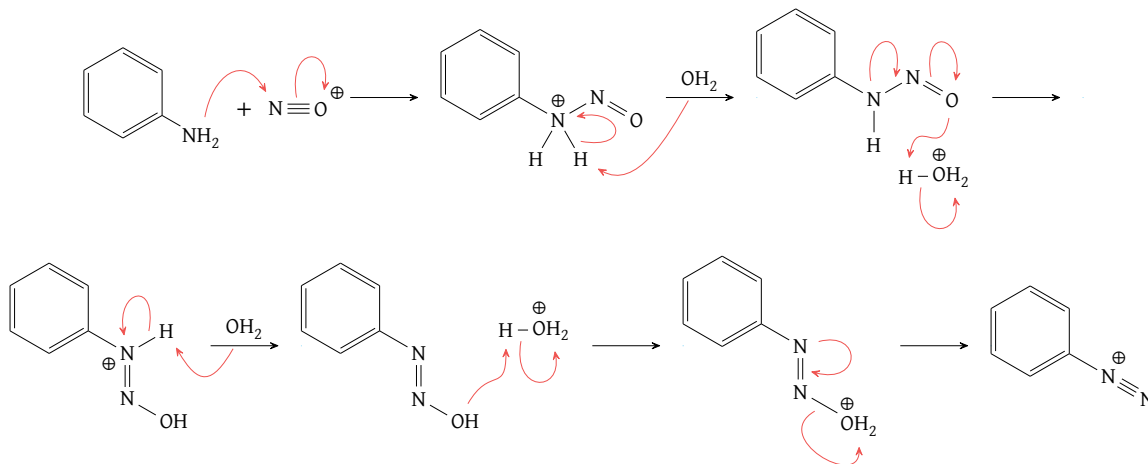


Figure 50: Conversion of aniline to aryl diazonium via interaction with the nitrosyl cation